

Generating Poetic Texts using RNN

In []:

In [30]:

```
#Import required libraries:
import random
import numpy as np
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Input
from tensorflow.keras.optimizers import RMSprop
```

In [31]:

```
#Get the Shakespeare text:
#https://storage.googleapis.com/download.tensorflow.org/data/shakespeare.txt
filepath = tf.keras.utils.get_file("shakespeare.txt", "https://storage.googleapis.com/download.tensorflow.org/data/shakespeare.txt")
```

In [32]:

```
#Get the text of this file into the script:
text = open(filepath, "rb").read().decode(encoding="utf-8").lower()
```

In [33]:

```
#Convert this text into a numerical-format (so that our neural-network is able to properly deal with the text):
#We will pass using the format of numpy array for our neural-network:

#We will only use a part of text (so that our neural-network doesn't take too much time on training):
text = text[300000:800000]

#Get a set of possible characters from the text:
characters = sorted(set(text))

#Create 2 dictionaries so that we can convert these characters from characters-to-numeric and back from numeric-to-character format:
#NOTE: The numerical values will represent the frequencies of each character.
char_to_index = dict((c,i) for i, c in enumerate(characters))
index_to_char = dict((i,c) for i, c in enumerate(characters))
```

In [34]:

```
#Get the feature-data and the target-data:
SEQ_LENGTH = 40
STEP_SIZE = 3

sentences = [] #For loading the sentence
next_characters = [] #for predicting the next character of the sentence
#for-example: 'how are yo'---->'u'----->'how are you'

for i in range(0, len(text)-SEQ_LENGTH, STEP_SIZE):
    sentences.append(text[i:i+SEQ_LENGTH])
    next_characters.append(text[i+SEQ_LENGTH])
```

In [35]:

```
#Convert the training-data into a numpy array:
x = np.zeros((len(sentences), SEQ_LENGTH, len(characters)), dtype=bool)
y = np.zeros((len(sentences), len(characters)), dtype=bool)
```

In [36]:

```
#Now fill these arrays:
for i, sentence in enumerate(sentences):
    for t, character in enumerate(sentence):
        x[i,t,char_to_index[character]] = 1
    y[i,char_to_index[next_characters[i]]] = 1
```

In [37]:

```
#Now build the neural-network:
model = Sequential()
model.add(Input(shape=(SEQ_LENGTH, len(characters))))
model.add(LSTM(128))
model.add(Dense(len(characters)))
model.add(Activation('softmax')) #activation layer with activation-function as SOFTMAX

#Compile the model along with the loss-function:
model.compile(loss="categorical_crossentropy", optimizer=RMSprop(learning_rate=0.01))

#Now fit the training-data:
model.fit(x,y,batch_size=256,epochs=4)

#Save the model:
model.save("textgenerator.h5")
```

```
Epoch 1/4
651/651  135s 204ms/step - loss: 2.1789
Epoch 2/4
651/651  131s 202ms/step - loss: 1.7391
Epoch 3/4
651/651  132s 203ms/step - loss: 1.6068
Epoch 4/4
651/651  131s 201ms/step - loss: 1.5337
```

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my\_model.keras')` or `keras.saving.save\_model(model, 'my\_model.keras')`.

In [38]:

```
'''
For loading the model:

from tensorflow.keras.models import load_model
model = load_model("textgenerator.keras")
'''
```

Out[38]:

```
'\nFor loading the model:\n\nfrom tensorflow.keras.models import load_model\nmodel = load_model("textgenerator.keras")\n'
```

In [39]:

```
#Function to predict the next character in the sentence:
def sample(preds, temperature=1.0):
    preds = np.asarray(preds).astype("float64")
    preds = np.log(preds) / temperature
    exp_preds = np.exp(preds)
    preds = exp_preds / np.sum(exp_preds)
    probas = np.random.multinomial(1, preds, 1)
    return np.argmax(probas)

#Function which generates text:
def generate_text(length, temperature):
    start_index = random.randint(0, len(text)-SEQ_LENGTH - 1)
    generated = ""
    sentence= text[start_index:start_index+SEQ_LENGTH]
    generated += sentence
```

```

for i in range(length):
    x = np.zeros((1, SEQ_LENGTH, len(characters)))
    for t, character in enumerate(sentence):
        x[0,t,char_to_index[character]] = 1

    predictions = model.predict(x, verbose=0)[0]
    next_index = sample(predictions, temperature)
    next_character = index_to_char[next_index]

    generated += next_character
    sentence = sentence[1:] + next_character

return generated

```

In [40]:

```

print("\n\n-----Temperature as 0.2-----")
print(generate_text(300,0.2))
print("\n\n-----Temperature as 0.4-----")
print(generate_text(300,0.4))
print("\n\n-----Temperature as 0.6-----")
print(generate_text(300,0.6))
print("\n\n-----Temperature as 0.8-----")
print(generate_text(300,0.8))
print("\n\n-----Temperature as 1-----")
print(generate_text(300,1))

```

-----Temperature as 0.2-----
where now we mean to stand.

second keeper:
the truth and many the cannot son i have in the son in the death.

second witchilings:
if you have the lanch the lanch and the counders
to the battle the crown and the king here with the beath,
i have the bears and the lanch the canst the dead;
and in the soul is the canst the come of the brothe

-----Temperature as 0.4-----
do it?

queen margaret:
ay, but thou use the summer the have in the brother heart:
where is the bears is shall be to but the trainor
to stand and speak of my lord of the counders,
i cannot the proposon to be our speak to the must.

nurse:
these propost the world a bear the see me son it is not say.

king richard ii:
why, which the burded

-----Temperature as 0.6-----
en neptune
a ram, and bleated; and the from that hearth.

rich of york:
why, hears of the battle be nuch is the blood,
and a why son i know to the brother as gone?

duke of gvester:
and maday, and for his in many the brow.
why, and engramen you she well you have dury at the world,
with this lancome, the master some string.

duke of aumerl

-----Temperature as 0.8-----
which he justly owes him.

archidamus:
plantiss father, till the king heart are ancenes
on ad, if you to diswands to the take.

matimet:
eye me; then be mostly of the bogat england.

peting:

romeo:

butnet:
i way lengted, his thou secrake the crown.

king richards:
not and he shear the lance his beathed presence
crome of all you tame m

-----Temperature as 1-----
e that thou art poor:
hold, there is for a husk and , set it speen,
men anage to 'eting propormens, untring.

undello?
nafule:
from nor a frow. that hold thy los kneel-vishort!

catulion:
man, shall come i, confoun bod: you way to
think her? the crep, no would he wash we fror oh name.

edward:

duke of murchet:

due-therchy of york:
and i

In []: